

A time-efficient CFD approach for hydrodynamic coefficient determination and model simplification of submarine

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ABSTRACT

Maneuverability is one of the most important performance characteristics of submarines. Hydrodynamic model is increasingly relied on as a design approach to determine the inherent motion behavior of a proposed submarine before construction. Standard submarine motion equations are the most used hydrodynamic model, in which more than 100 coefficients need to be estimated. These coefficients have to be determined by captive model tests, semi-empirical method and potential flow method on the basis of existing approach. The separately determining approach makes it difficult to assess model reliability. This paper proposes a time-efficient approach for the estimation of hydrodynamic coefficients using computational fluid dynamics (CFD) method. Instead of a repetitive and time-consuming process, the proposed spatial captive motion could provide all necessary information for determination of all required coefficients in only one simulation. Linear regression, based on the least square method, is employed to determine coefficients in hydrodynamic model. Statistical investigation, correlation and significance analysis, indicates that the standard submarine motion equations can be further simplified when submarine moving forward with small perturbations along other directions. The final simplified motion equations have much fewer components than the original model while fitting accuracy remained. Validation results prove that spatial captive motion simulation and the simplification approach employed in this paper are effective and reliable. It should have a large scope in further maneuverability research.

1. Introduction

The study of maneuvering characteristics have received considerable attention in the past and is becoming increasingly important in the design and operation of submarine. This helps determine motion characteristics prior to the construction of vehicles and remedy any shortcomings at the redesign stage. A carefully designed submarine should have good performance in both motion stability and maneuverability, which are conflicted but interrelated. Most maneuvering characteristics studies are based on hydrodynamic motion equations. Looking at the commonly used standard submarine motion equations (Gertler and Hagen., 1967), as many as 100 coefficients need to be estimated. If the 2nd Taylor expansion method is employed, the number would be over 200. An efficient and correct method for coefficient prediction is essential for the study of maneuvering characteristics.

Free-running test is the best approach for studying maneuvering characteristics (Dantas et al., 2013), but the cost is quite high, and the test accuracy could be easily affected by the experimental environment

and measuring instruments. Captive-model experiments have been widely used to predict coefficients since the 1950s. Planar motion mechanism (PMM) (Goodman and Gertler., 1962) and rotating arm (RA) (Burcher, 1969) are the most common methods, RA method could be used to determine the values of coefficients that represent the forces and moments as functions of yaw and pitch velocity. It could also be used to obtain some values of cross coefficients related to angular velocity, but it is difficult to obtain values of sway, heave and most of the other cross coefficients. Another reason that the RA method is not widely applied is that it demands more experimental facilities and higher construction cost. On the contrary, the PMM costs less and can determine more information. It could be used to simulate pure translation or rotation, which would obtain the added masses, values of coefficients that represent the forces and moments as functions of heave, pitch, sway, and yaw. Combined with empirical formulas, all the coefficients required for the submarine motion equations can be calculated.

Captive-model experiment was, is, and will be in the predictable future, maintaining its primary role in coefficient estimation, but the test

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plan and uncertainty analysis have to be carefully considered to make it more reliable (Sutulo and Kim., 1998). With improvements in computer technology, computational fluid dynamics (CFD) is a fast moving field in hydrodynamic performance prediction of naval architecture (Stern et al., 2013, 2015) owing to its high efficiency and low cost. To date, CFD has been widely used to simulate PMM test (Kim et al., 2015), RA test (Wu et al., 2015), or other model tests (Mansoorzadeh and Javanmard., 2014; Randeni et al., 2015; Singh et al., 2017) to obtain the required coefficient values or to study hydrodynamic performance. There has also been considerable study on the application of Reynolds–Averaged Naviers Stokes (RANS) formulation to the hydrodynamic performance analysis of submarine and other naval architectures (Araki et al., 2012; Fureby et al., 2016; Mansoorzadeh and Javanmard., 2014; Motoki et al., 2013; Pan et al., 2012; Toxopeus et al., 2012). Another CFD developing area is free maneuvering simulation (Carrica et al., 2010; Chase et al., 2013; Shen et al., 2015). Those works prove that CFD is continuing advance at ever increasing speed and ever-more-impressive achievements.

As remarkable as model tests and CFD development are, thus far, the obtaining methods for hydrodynamic coefficients are still restricted to the PMM test, RA test and related CFD simulations. The investigation of coefficients for submarine only pays close attention to several primary factors. There is still not an efficient approach to estimate all the required coefficients in submarine motion equations, not to mention equation reliability assessment and coefficients' significance tests. In this study, spatial captive motion using a CFD approach is developed to estimate all the hydrodynamic coefficients except body-rudder-propeller interaction related coefficients.

The paper is organized as follows. Section 2 introduces the main concepts and numerical methods. Section 3 describes the hydrodynamic coefficient estimation method on the basis of spatial captive motion simulation using a CFD approach. Section 4 discusses the equation accuracy and coefficients significance based on statistical method. Section 5 presents further experimental verification of the proposed approach. Section 6 reviews the achieved progress and proposes future developments for this study.

Formulations presented in this paper closely follow that carried out for submarine as described by Gertler Hagen (1967). Notations used for the formulation are as follows.

f_i	Frequency
K, M, N	Moments about the x-axis, y-axis and z-axis respectively
K', M', N'	Non-dimensional moments about the x-axis, y-axis and z-axis respectively, for example $K' = K / \left(\frac{1}{2} \rho V^2 l^3 \right)$
$K'_p, M'_q, N'_r, \dots$	Non-dimensional coefficient of moments due to different velocity components
$K''_p, M''_q, N''_r, \dots$	Non-dimensional coefficient of moments due to different acceleration components
l	Characteristic length for non-dimensionalization of the hydrodynamic forces and moments
$O - \xi\eta\zeta$	Earth-fixed and body-fixed coordinates
$O - xyz$	
p, q, r	Angular velocities about the x-axis, y-axis and z-axis respectively
$\dot{p}, \dot{q}, \dot{r}$	Angular accelerations about the x-axis, y-axis and z-axis respectively
r_{pec}	Pearson Correlation Coefficient
t	Time
u, v, w	Velocities in the x, y and z directions respectively
$\dot{u}, \dot{v}, \dot{w}$	Accelerations in the x, y and z directions respectively
V, V_0, V_p	Velocity, constant velocity, range of velocity perturbation
X, Y, Z	Forces in the x-axis, y-axis and z-axis respectively
X', Y', Z'	Non-dimensional forces in the x-axis, y-axis and z-axis respectively, for example $X' = X / \left(\frac{1}{2} \rho V^2 l^2 \right)$
$X'_{us}, Y'_{vs}, Z'_{ws}, \dots$	Non-dimensional coefficient of forces due to different velocity components
$X'_{\dot{u}}, Y'_{\dot{v}}, Z'_{\dot{w}}, \dots$	Non-dimensional coefficient of forces due to different acceleration components
y^+	Dimensionless wall distance
ρ	Fluid density
Δy	the distance of first layer from the body surface

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(continued)

μ, ν	dynamic and kinematic viscosity coefficient respectively
μ_τ	the friction velocity
τ_w	the shear stress at the body surface

2. Mathematical model

Motion can be described by the combination of two coordinate systems, one is the Earth-fixed reference coordinate system, and the other is the body-fixed coordinate system. As shown in Fig. 1, translations and rotations of submarine are defined with respect to the Earth-fixed coordinate system, where linear and angular velocity, and forces and moments, are defined with respect to the body-fixed coordinate system.

For a maneuvering submarine, the relationship between motion variables and the hydrodynamic forces and moments can be represented by a mathematical model in terms of a series of coefficients. Standard submarine motion equations are commonly used in the submarine community as a typical mathematical model to represent the hydrodynamic forces and moments with respect to the body fixed coordinate system. When rudder rotation and propeller performance are not considered, hydrodynamic forces and moments can be defined as follows:

$$X = \frac{1}{2} \rho l^4 (X'_{qq} q^2 + X'_{rr} r^2 + X'_{rp} rp) + \frac{1}{2} \rho l^3 (X'_{\dot{u}} \dot{u} + X'_{vr} vr + X'_{wq} wq) + \frac{1}{2} \rho l^2 (X'_{uu} u^2 + X'_{vv} v^2 + X'_{ww} w^2) \quad (1)$$

$$Y = \frac{1}{2} \rho l^4 (Y'_{\dot{r}} \dot{r} + Y'_{\dot{p}} \dot{p} + Y'_{p|p|} p|p| + Y'_{pq} pq + Y'_{qr} qr) + \frac{1}{2} \rho l^3 \left(Y'_{\dot{v}} \dot{v} + Y'_{vq} vq + Y'_{wp} wp + Y'_{wr} wr + Y'_{ur} ur + Y'_{up} up + Y'_{v|v|} \frac{v}{|v|} \left| (v^2 + w^2)^{\frac{1}{2}} \right| |r| \right) + \frac{1}{2} \rho l^2 \left(Y'_{\dot{u}} \dot{u}^2 + Y'_{uv} uv + Y'_{v|v|} v \left| (v^2 + w^2)^{\frac{1}{2}} \right| + Y'_{vw} vw \right) \quad (2)$$

$$Z = \frac{1}{2} \rho l^4 (Z'_{\dot{q}} \dot{q} + Z'_{pp} p^2 + Z'_{rr} r^2 + Z'_{rp} rp) + \frac{1}{2} \rho l^3 \left(Z'_{\dot{w}} \dot{w} + Z'_{vr} vr + Z'_{vp} vp + Z'_{qu} qu + Z'_{w|q|} \frac{w}{|w|} \left| (v^2 + w^2)^{\frac{1}{2}} \right| |q| \right) + \frac{1}{2} \rho l^2 \left(Z'_{\dot{u}} \dot{u}^2 + Z'_{uv} uv + Z'_{uw} uw + Z'_{w|w|} w \left| (v^2 + w^2)^{\frac{1}{2}} \right| + Z'_{|w|} w |w| + Z'_{ww} w (v^2 + w^2)^{\frac{1}{2}} + Z'_{vv} v^2 \right) \quad (3)$$

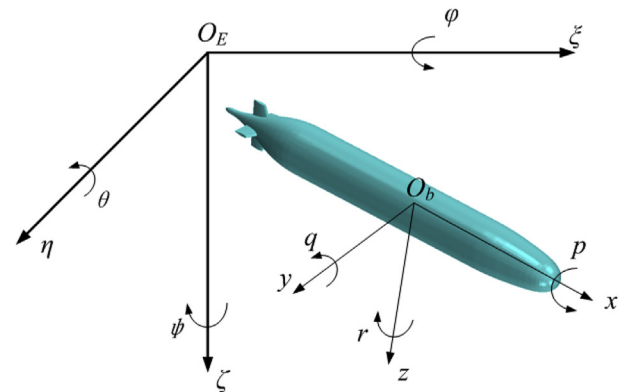


Fig. 1. Description of the coordinate systems.

$$K = \frac{1}{2}\rho l^5 (K'_{\dot{p}\dot{p}} + K'_{\dot{r}\dot{r}} + K'_{qr}qr + K'_{pq}pq + K'_{p|p|}p|p|) + \frac{1}{2}\rho l^4 (K'_{p\dot{u}p} + K'_{r\dot{u}r} + K'_{\dot{v}\dot{v}} + K'_{vq}vq + K'_{wp}wp + K'_{wr}wr) + \frac{1}{2}\rho l^3 \left(K'_{*u^2} + K'_{*uv} + K'_{*v|v|}v \left| (v^2 + w^2)^{\frac{1}{2}} \right| + K'_{*vw}vw \right) \quad (4)$$

$$M = \frac{1}{2}\rho l^5 (M'_{\dot{q}\dot{q}} + M'_{pp}p^2 + M'_{rr}r^2 + M'_{rp}rp + M'_{q|q|}q|q|) + \frac{1}{2}\rho l^4 \left(M'_{*w\dot{w}} + M'_{*vr}vr + M'_{*vp}vp + M'_{*uq}uq + M'_{*v|v|}v \left| (v^2 + w^2)^{\frac{1}{2}} \right| q \right) + \frac{1}{2}\rho l^3 \left(M'_{*u^2} + M'_{*uw}uw + M'_{*w|w|}w \left| (v^2 + w^2)^{\frac{1}{2}} \right| + M'_{*w|u|}w|u| + M'_{*ww}w \left| (v^2 + w^2)^{\frac{1}{2}} \right| + M'_{*vw}v^2 \right) \quad (5)$$

$$N = \frac{1}{2}\rho l^5 (N'_{\dot{r}\dot{r}} + N'_{\dot{p}\dot{p}} + N'_{pq}pq + N'_{qr}qr + N'_{r|r|}r|r|) + \frac{1}{2}\rho l^4 \left(N'_{*v\dot{v}} + N'_{*wr}wr + N'_{*wp}wp + N'_{*vq}vq + N'_{*p\dot{u}p} + N'_{*r\dot{u}r} + N'_{*v|r|}v \left| (v^2 + w^2)^{\frac{1}{2}} \right| r \right) + \frac{1}{2}\rho l^3 \left(N'_{*u^2} + N'_{*uv}uv + N'_{*v|v|}v \left| (v^2 + w^2)^{\frac{1}{2}} \right| + N'_{*vw}vw \right) \quad (6)$$

Estimation of hydrodynamic coefficients is a key step to predict maneuverability. The most common way to determine the value of those coefficients is captive model tests including the oblique towing tests, the RA tests and the PMM tests. However, the PMM model tests or CFD simulation could determine only fewer than 20 hydrodynamic coefficients. Besides, to get more reliable coefficients, repetitive tests or simulations with different velocities or frequencies have to be conducted to get their mean values. In addition, the other coefficients have to be obtained using semi-empirical method and potential flow method. But the semi-empirical method cannot take the complete hydrodynamic arrangement into account, and the potential flow method can only determine inertial hydrodynamic coefficients without consideration of water viscosity. The inaccuracy of semi-empirical method and potential flow method would also seriously impair the equation's reliability. Furthermore, reliability of mathematical model cannot be ensured when hydrodynamic coefficients are determined separately by different approaches.

3. Spatial captive motion simulation

To ensure reliability of mathematical model, it would be best if all hydrodynamic coefficients can be determined by one approach and one simulation. This can be reached if submarine moves irregularly in every direction while forces and moments being monitored at all time. It could not be conducted in free-running test and towing tank test. This study proposes a novel approach, spatial captive motion simulation in CFD, to predict all the hydrodynamic coefficients in mathematical model by only one simulation.

3.1. Preliminary tests

Numerical simulation is performed with the commercial CFD code, STAR-CCM+, using the finite volume, viscous and incompressible unsteady RANS equations. The $k-\omega$ shear stress transport (SST) turbulence model is employed owing to its good prediction capability with reasonable computational cost. Works by Lloyd and Espanoles (2002), Jak and Hayes (2008), Dantas and Barros (2013) are taken as reference to guide design of simulation strategy. Grids of up to 16.4 million cells have been generated for DARPA SUBOFF, whose geometric characteristics (Groves et al., 1989) as shown in Fig. 2, in straight-ahead to balance well between computational cost and simulation accuracy. Meshes are generated using the trimmed mesher embedded in STAR-CCM+. The trimmed mesher generates 3D meshes containing a predominantly hexahedral mesh with minimal skewed cell. User-defined refinement can be done easily in any region. It is quite an efficient and robust method of generating high quality grids for complex geometries. The mesh size is increased from the SUBOFF body surface to the outer region.

Turbulence models require a reasonable thickness for the volume mesh adjacent to the body surface. It is defined as a dimensionless wall distance y^+ , which is given as

$$y^+ = \frac{\Delta y \rho \mu_\tau}{\mu} = \frac{\Delta y}{\nu} \sqrt{\frac{\tau_\omega}{\rho}} \quad (7)$$

All- y^+ wall treatment is applied to the solver. It is a hybrid treatment, for coarse meshes, y^+ is high and the near-wall cell lies within logarithmic region. While for fine meshes, y^+ is low enough to ensure that wall cells are within the buffer region, a sub layer could be properly resolved using low-Reynolds number turbulence models.

Fig. 3 shows the mesh around the SUBOFF generated by the trimmed mesher.

Unsteady numerical simulations of the SUBOFF in a straight-ahead direction are performed. Differing from the frequently used method that converts an unsteady motion problem to a steady-state problem, the work in this study involves actual displacement of mesh vertices in real time. Simulations with grid sizes of up to 16.4 million cells are performed to determine the optimal mesh strategy to improve the simulation accuracy. The straight-ahead direction simulation at velocities of 5.92kn is simulated in STAR-CCM+. The drag measured by experiment is 95.41N at a velocity of 5.92kn which corresponds to a Reynolds number of about 1.4×10^6 (Liu and Huang., 1998). Table 1 shows the relative error variation while increasing the mesh size. When the mesh is coarse, the simulation results are not entirely reliable. The simulation accuracy does not improve when the mesh size is above a certain level. To ensure simulation accuracy and reduce computation costs, the grid size of 4.96 million is adopted to conduct the follow-up research. The relative error would be within 5%.

According to the preliminary investigations and test results, the simulation strategy including viscous and incompressible unsteady RANS equations, $k-\omega$ SST turbulence model, the all- y^+ treatment, trimmed mesher with user-defined refinement is confirmed as a guideline for all simulations. Its validity is further proved in the subsequent comparisons between CFD and EFD results.

3.2. Spatial captive motion simulation

The relationship between motion variables and the hydrodynamic forces and moments can be taken as a model under consideration, which can be written in the form

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad (8)$$

where $\mathbf{Y}$ is the vector of hydrodynamic forces or moments, $\mathbf{X}$ is a matrix

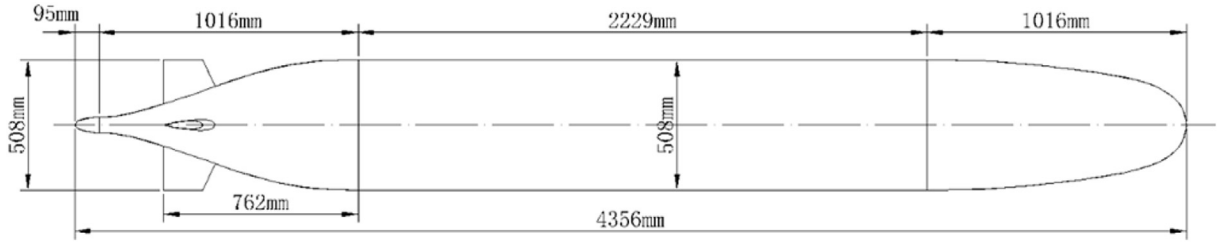


Fig. 2. Geometry details of the DRAPA SUBOFF.

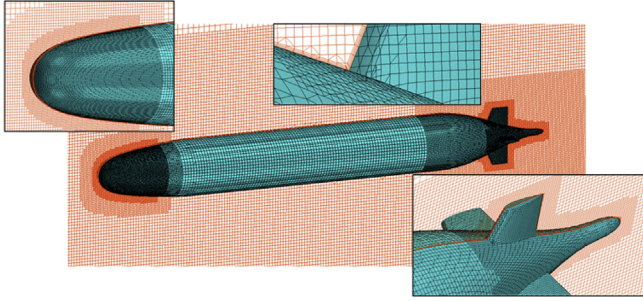


Fig. 3. Trimmed mesh.

Table 1
Comparison between computational and experimental value for the SUBOFF model.

Mesh size (million cells)	Average y + value	Computational value (N)	Experimental value (N)	Relative error
0.48	98.41	102.87	95.41	7.82%
2.07	38.66	89.01		−6.71%
4.96	27.9	91.43		−4.17%
7.48	24.1	92.01		−3.56%
16.35	18.47	93.67		−1.82%

of known form of motion variables, β is a vector of parameters, and ϵ is a vector of errors. If enough data information of motion variables and hydrodynamic forces or moments can be monitored in one simulation, the vector of parameters can then be determined. To address this, the supposed time-efficient simulation should have two essential features, one is that velocities and accelerations, $[u, v, w, p, q, r, \dot{u}, \dot{v}, \dot{w}, \dot{p}, \dot{q}, \dot{r}]$ are all gradually but constantly changing, the other is that all motion information and their corresponding hydrodynamic forces and moments are monitored.

Motion definition is important to reach reasonable hydrodynamic coefficients. It should be as similar to motion in actual ocean environment as possible. Considering the fact that hydrodynamic coefficients are calculated on the basis of steady state assumption, motion in a straight-ahead direction with artificial random and smoothing perturbations is simulated in STARCCM+. For linear and angular velocities with respect to the body-fixed coordinate system, u, v, w, p, q, r , they can be defined as

$$V = V_0 + V_p \sin(2\pi f_1 t) \cdot \sin(2\pi f_2 t) \cdot \sin(2\pi f_3 t) \quad (9)$$

For the translation velocity along x-axis, V_0 should be cruising speed around which hydrodynamic coefficients would be calculated, for other velocities, V_0 should be zero. To ensure coefficients identification available, values of f_i are well determined to make all velocities fully unrelated to each other. Forces and moments with respect to the body-fixed coordinate system are monitored and recorded. Fig. 4 shows the velocities and forces variation between time 280s and 300s.

3.3. Coefficient identification

The mathematical model can be taken as a predictive model to an observed data set of hydrodynamic forces and velocities values. Taking sway force Y as an example, its mathematical model can be written as

$$Y_i = \beta_1 Y_{1,i} + \beta_2 Y_{2,i} + \dots + \beta_k Y_{k,i} + \epsilon_i \quad (10)$$

Where Y is the hydrodynamic sway force, Y_k are explanatory variables such as $\dot{r}, \dot{p}, p|p|, \beta_k$ are coefficients to be estimated, ϵ_i are errors between monitored value and predicted value. When such a model developed, if additional velocity values are given without its accompanying hydrodynamic forces, the fitted model can make a prediction of hydrodynamic forces. This is a typical problem of regression analysis, especially, as all coefficients are linear, hydrodynamic coefficients can be determined by linear regression. The core issue is to identify proper value of β_k to minimize the error ϵ_i . This objective can be reached by employing Ordinary Least Squares approach (Draper et al., 1998), which is conceptually simple and computationally straightforward.

A total time of 334s of spatial captive motion is simulated in STAR-CCM+. It takes 64.6 h on a work station which is a 64 bit processor consist of 2 CPUs (Intel Xeon E5-2640 @ 2.50 GHz) and 48 GB accessible memory. After an initial simulation of 15s, the numerical flow field converges, linear and angular velocities and accelerations $u, v, w, p, q, r, \dot{u}, \dot{v}, \dot{w}, \dot{p}, \dot{q}, \dot{r}$, and their corresponding forces and moments are then recorded every 0.2s. The data ranges from 15s to 270s are adopted to obtain the hydrodynamic coefficients using linear regression based on the least square method. The remaining data range from 320s to 330s is kept for validation. As shown in Fig. 5, with the identified coefficients by linear regression, the mathematical model shows perfect performance on accommodating all forces and moments.

To make the validation work more comparable, the coefficient of determination R -square is calculated.

$$R - square = \frac{SSR}{SST} \quad (11)$$

Where SSR is sum of squares of the regression,

$$SSR = \sum_{i=1}^n \omega_i (\hat{y}_i - \bar{y}_i)^2 \quad (12)$$

SST is the total sum of squares

$$SST = \sum_{i=1}^n \omega_i (y_i - \bar{y}_i)^2 \quad (13)$$

The closer to 1 the R -square value is, the more the equation fits the simulation result. According to the R -square values shown in Table 2, with the identified coefficients by linear regression, the mathematical model shows perfect performance on accommodating all forces and moments.

Table 3 demonstrates the required number of simulations to determine hydrodynamic coefficients in standard submarine motion

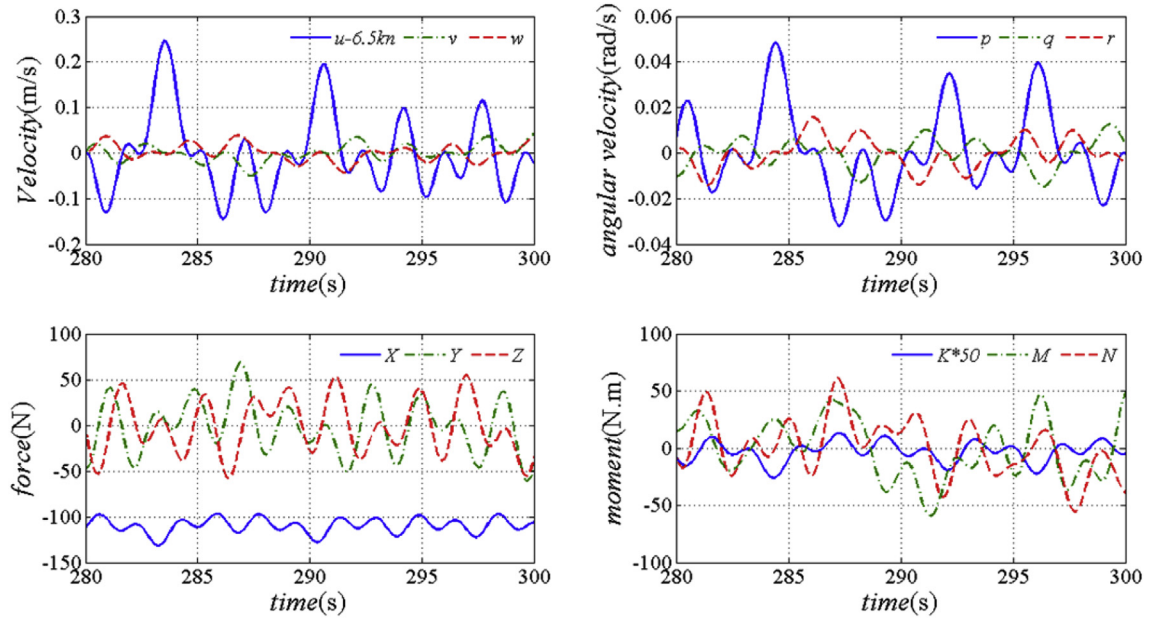


Fig. 4. Linear and angular velocity variations with perturbations.

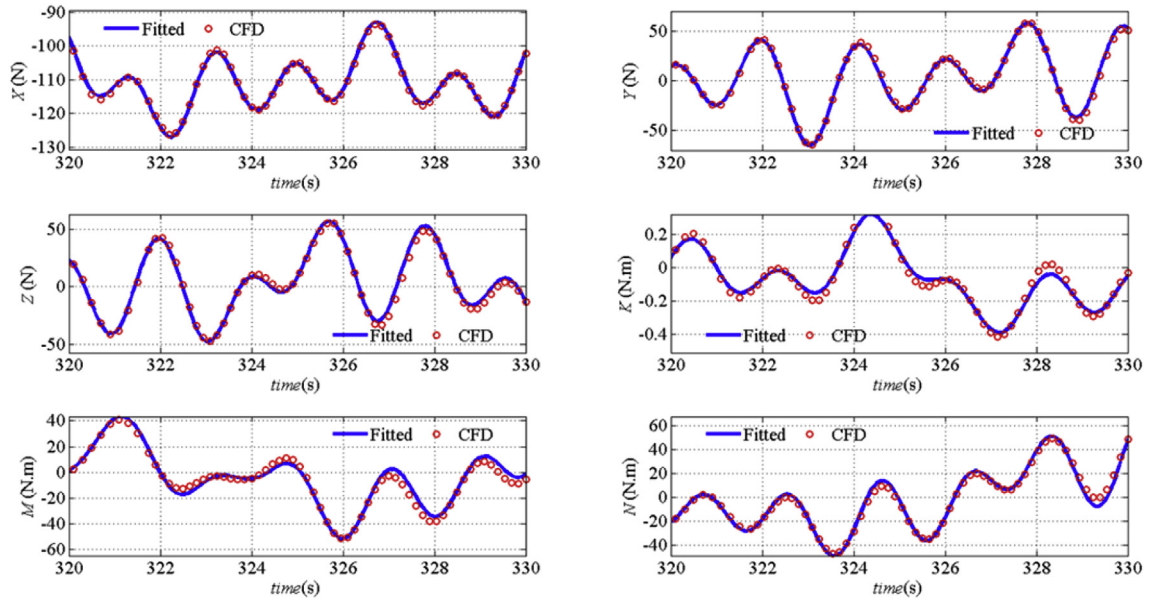


Fig. 5. Forces and moments fitting.

Table 2
R-square statistic.

Force/Moment	R-square
X	0.9953
Y	0.9915
Z	0.9902
K	0.9646
M	0.9875
N	0.9903

equations. Although repetitive simulations are required, PMM and RA can determine very limited coefficients. Spatial captive motion simulation, by contrast, determines all coefficients in equation (1)–(6) in less time. More importantly, it can be used for accuracy assessment of

hydrodynamic model. *R-square* values above 0.95 in all directions indicate that standard submarine motion equations are good enough to model SUBOFF moving forward with small perturbations along other directions.

4. Analysis of mathematical model

Spatial captive test simulation helps determine all hydrodynamic coefficients efficiently, but saving time is not its only advantage. Different from PMM and RA determining coefficients with repetitive simulations, the complete data set from spatial captive test simulation offers a probability of further research on mathematical model. Correlational analyses help investigate relationships between components and total forces. Significance test can figure out unimportant components. This can be applied to model simplification.

Table 3
Required simulations to determine coefficients in hydrodynamic model.

Coefficients	Pure surge	Pure Sway (heave) PMM	Pure Yaw (pitch) PMM	RA	Spacial captive motion
X'_{uu}	✓				✓
$X'_{\dot{u}}$	✓				✓
$Y'_v(Z'_w)$		✓			✓
$N'_v(M'_w)$		✓			✓
$Y'_v(Z'_w)$		✓			✓
$N'_v(M'_w)$		✓			✓
$Y'_r(Z'_q)$			✓	✓	✓
$N'_r(M'_q)$			✓	✓	✓
$Y'_r(Z'_q)$			✓	✓	✓
$N'_r(M'_q)$			✓	✓	✓
Coupling terms				Some	All

4.1. Investigation of hydrodynamic forces and moments

According to definition in equation (10), all equations of hydrodynamic forces and moments can fall into some component forces or moments due to that velocities or accelerations are variable. The correlation and dependence between components and total force can be quantified by Pearson correlation coefficient (PCC) (Nicewander, 1988), which is the covariance of the two variables divided by the product of their standard deviations. The PCC can be calculated by

$$r_{pcc} = \frac{\sum X_1 X_2 - \frac{\sum X_1 \sum X_2}{k}}{\sqrt{\left(\sum X_1^2 - \frac{(\sum X_1)^2}{k}\right) \left(\sum X_2^2 - \frac{(\sum X_2)^2}{k}\right)}} \quad (14)$$

r_{pcc} has a value between -1 and +1. It measures the linear dependence between two variables X_1 and X_2 . $r_{pcc} = 1$ means two variables are total positive linear correlation, and -1 means total negative linear correlation. If $r_{pcc} = 0$, the two variables should be no linear correlation.

Fig. 6 shows all values of r_{pcc} between each component force or moment and their corresponding total force or moment. The number

along the x-axis is the order according to equation (1)–(6). For the force in the longitudinal axis, X , it is highly positive linear correlated with $\dot{u}$ and u^2 , negative factors do not show as much correlation as positive ones; sway force Y and heave force Z are most positive linear correlated with linear accelerations along y-axis and z-axis, $\dot{v}$ and $\dot{w}$, respectively; roll moment K is highly positive linear correlated with the product of u and p , and negative linear correlated with $p|p|$; pitch moment M and yaw moment N are most positive linear correlated with linear velocities along z-axis and y-axis, w and v , respectively.

According to the analysis result about correlation, for over half of coefficients, their values of r_{pcc} with total force or moment are very closed to 0. It indicates that these component parts are uncorrelated to the total force or moment. Further investigation on their significances can be conducted. If very limited importance is shown in the total force or moment, the corresponding component part can be removed to simplify the original mathematical model.

4.2. Significance test and model simplification

There are up to 88 coefficients need to be estimated in the original hydrodynamic model. Their significances should be highly different from each other. Some coefficients may show so limited influence to the total forces that can be eliminated. This probability of eliminating can be evaluated by statistical hypothesis testing (Shao, 2003). It is defined as

$$H_0 : \beta_{ij} = 0 \quad (15)$$

Where, β_{ij} represents the j th coefficient of equation i . Null hypothesis H_0 means that the selected coefficient β_{ij} is not significant and more probability be equal to zero. The p -value is proposed as the smallest level of significance at which we would reject the null hypothesis H_0 . If the p -value is closer to zero, we come to the conclusion that the null hypothesis H_0 is false. Conversely, if the p -value is closer to one, it suggests more probability of the corresponding coefficient being zero.

The p -value is calculated to figure out the significance of all coefficients in motion equations as shown in Fig. 7. The closer to 1 the p -value is, the more probability that the null hypothesis of corresponding coefficient is satisfied. Among all coefficients, removing the parameter

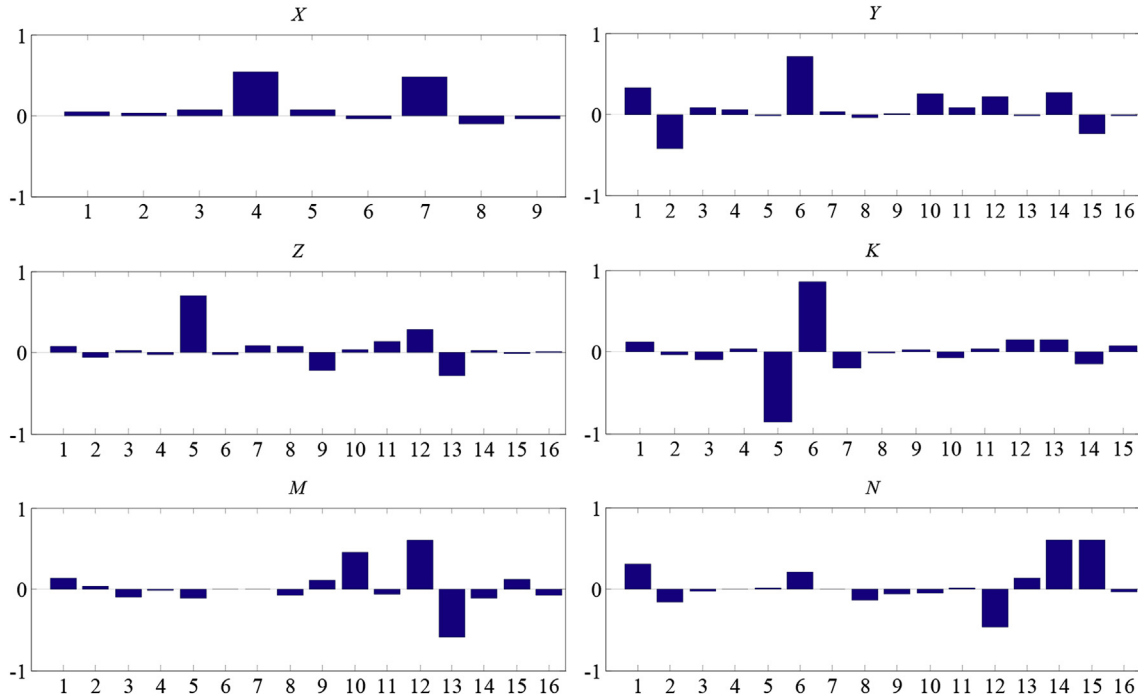


Fig. 6. Values of r_{pcc} between each component force/moment and total force/moment.

whose corresponding p -value is the highest will lead to the least impact on model accuracy.

On the basis of motion results from spacial captive motion simulation, observation matrix and regression matrix can be reached, and all coefficients can be determined using the previous introduced regression method. Then the significance test is conducted to figure out and eliminate unimportant coefficients. Finally, a new simplified dynamic model with remained accuracy and less coefficients is acquired. Specific method information is depicted in Fig. 8. While simplifying the mathematical model, 80 percent of the data are adopted to determine hydrodynamic coefficients, and the rest are kept as cross-validation study data for checking the model accuracy, which is evaluated by the value of R -square. If the value of R -square shows no obvious drop and maintains above 0.95, the coefficient with the least significance, i.e. with the highest p -value, is to be figured out and removed. The updated hydrodynamic model is then applied for parameters identification and model simplification. The iteration continues until model accuracy drops obviously. The final optimal hydrodynamic model is then determined.

Fig. 9 shows the R -square variations when simplifying the motion equations. As omitting coefficients, the values of R -square decrease very limited and maintain above 0.95 until certain iteration, then they suddenly drop below 0.95.

For the SUBOFF motion in this paper, the final simplified hydrodynamic model with the fewest coefficients and R -square above 0.95 are given as:

$$X = \frac{1}{2}\rho l^3 X'_{\dot{u}}\dot{u} + \frac{1}{2}\rho l^2 X'_{uu}u^2 \quad (16)$$

$$Y = \frac{1}{2}\rho l^3 Y'_{\dot{v}}\dot{v} + \frac{1}{2}\rho l^3 Y'_{ur}ur + \frac{1}{2}\rho l^2 Y'_{uv}uv \quad (17)$$

$$Z = \frac{1}{2}\rho l^3 Z'_{\dot{w}}\dot{w} + \frac{1}{2}\rho l^3 Z'_{uq}uq + \frac{1}{2}\rho l^2 Z'_{uw}uw \quad (18)$$

$$K = \frac{1}{2}\rho l^5 K'_{\dot{p}}\dot{p} + \frac{1}{2}\rho l^4 K'_{pu}pu + \frac{1}{2}\rho l^4 K'_{\dot{v}}\dot{v} + \frac{1}{2}\rho l^3 K'_{*}u^2 + \frac{1}{2}\rho l^2 K'_{uv}uv \quad (19)$$

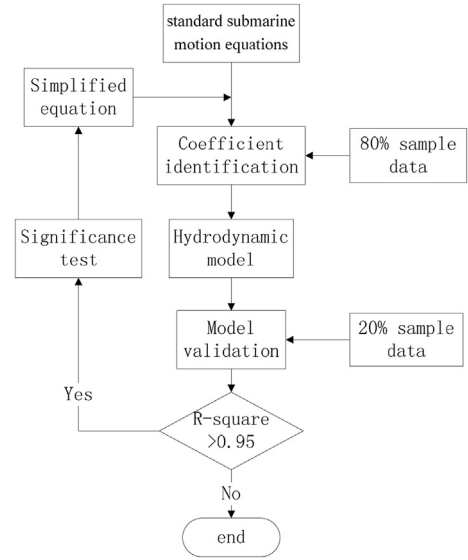


Fig. 8. Flow chart of model simplification process.

$$M = \frac{1}{2}\rho l^5 M'_{\dot{q}}\dot{q} + \frac{1}{2}\rho l^4 M'_{qu}qu + \frac{1}{2}\rho l^3 M'_{uw}uw \quad (20)$$

$$N = \frac{1}{2}\rho l^5 N'_{\dot{r}}\dot{r} + \frac{1}{2}\rho l^4 N'_{ur}ur + \frac{1}{2}\rho l^3 N'_{uv}uv \quad (21)$$

4.3. Fitting comparison

The high R -square value indicates that standard submarine motion equations are capable of describing properties of forces and moments acting on vehicle while moving forward with low oscillation. Furthermore, according to the analysis result of correlation and significance, some components in these equations have so limited impact on prediction of submarine performance that they could be removed. Specifically, after reducing 69 parameters, the final simplified hydrodynamic model

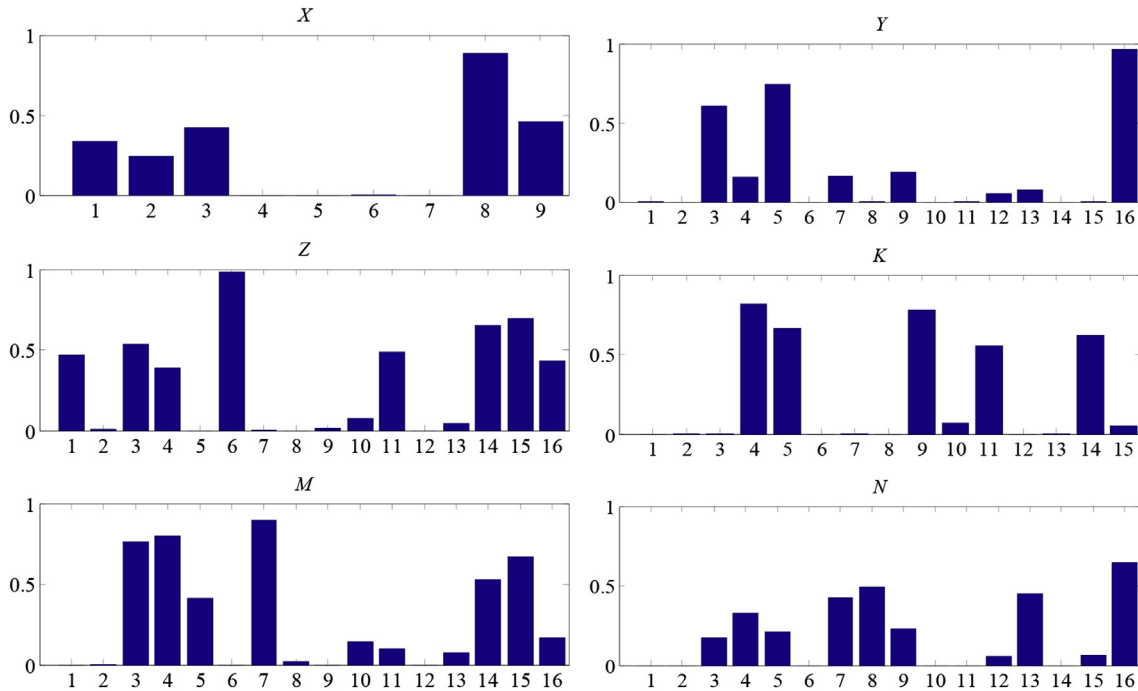


Fig. 7. P-value of coefficients.

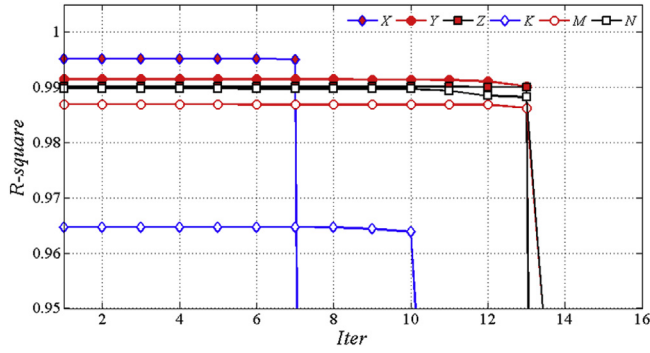


Fig. 9. R-square value variation with Equation simplification procedure for SUBOFF.

with only 19 parameters also shows perfect fitting performance with values of *R-square* above 0.95. To test the effectiveness of the simplified model and verify the ability of describing the change in forces and moments of CFD results, a set of analyses are performed to compare with EFD results by David Taylor Research Center (DTRC) (Roddy, 1990). Table 4 reveals comparison of coefficients for the horizontal plane between CFD results by the approach employed in this paper and EFD results by DTRC. CFD results achieve reasonable agreement with EFD results for most cases. Fig. 10(a) shows sway force *Y* and yaw moment *N* obtained by CFD approach, and their fitting curves according to hydrodynamic coefficients by CFD and EFD, respectively. In addition, the proportion of each coefficients on related force and moment equations are components analysis (see Fig. 10(b) and (c)). As shown in figures, the weight of each coefficient on the EFD and CFD systems are almost the same. In addition, Y_r' and N_v' which are removed from simplified equations also have the lowest share of sway force and yaw moment, respectively, and trending to zero in EFD curves. It can be figured out that, despite with errors existing in Table 4, the curves of force and moment have the same changing trend and high agreement with each other. Above all, the proposed method could determine all required coefficients and their significance to corresponding forces and moments in only one simulation. Fitting comparison results suggest that the novel coefficient determination and model simplification method in this paper is efficient and reasonably reliable.

5. Verification and validation

A submarine model constructed by Harbin Engineering University (HEU) is applied to further validate the coefficient determination approach presented in this paper. The model is designed referring to many well-known submarines, such as Dragon of Japanese, U212A of Germany, etc. The overall length of the model is 3.08 m. It has a sail at bow; two control surfaces symmetrically arranged both in horizontal and vertical planes aft. The shape of its cross section is NACA0012.

5.1. Experiment setup

PMM tests were conducted to determine hydrodynamic coefficients of

Table 4
Hydrodynamic coefficients of SUBOFF.

Coefficients	CFD	EFD	Error
Y_v'	-0.017323	-0.014711	-17.8%
Y_r'	0.005830	0.006324	-7.8%
Y_p'	-0.015000	-0.010494	-42.9%
Y_f'	-	0.000465	-
N_r'	-0.003095	-0.003064	-1.0%
N_p'	-0.000866	-0.000744	-16.3%
N_v'	-0.009720	-0.011254	13.6%
N_f'	-	0.000415	-

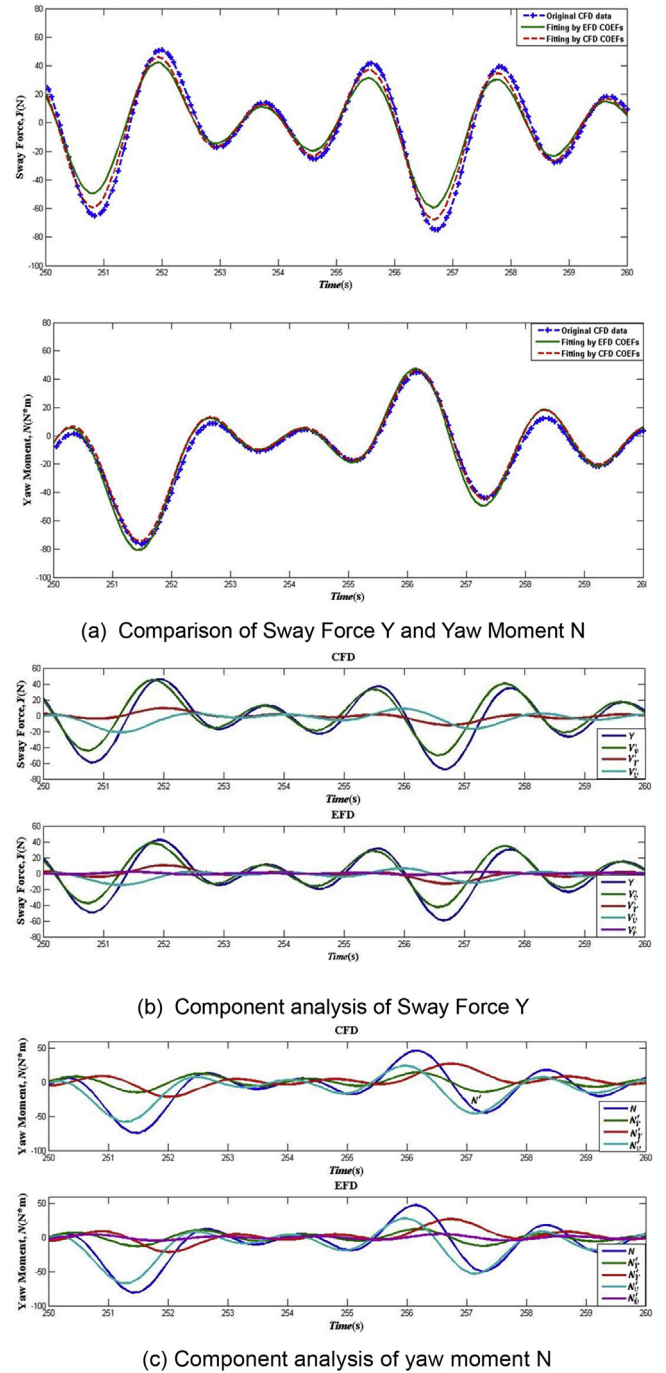


Fig. 10. Forces and Moments comparison in horizontal plane.

the HEU submarine model. A series of PMM tests, pitching, swaying, heaving, yawing and rolling, were performed in the circulating water tunnel of the State Key Laboratory of Science and Technology on Autonomous Underwater Vehicles at Harbin Engineering University, China. The circulating water tunnel, as shown in Fig. 11, is 7 m in effective length, 1.7 m in width and 1.5 m in depth, with a steady velocity range of 0.2 m/s to 2.0 m/s. A six-component balance was fixed in the submarine model. Two rapiers were used to connect to the balance and hold the model still while water flowed past it. The body was submerged in water of a depth three times the body diameter to minimize errors due to free surface. As a further measure to minimize other effects on the force and moment measurements, diameter of rapiers was tapered as small as possible, 10 mm, near the model surface. Fig. 11 shows the HEU

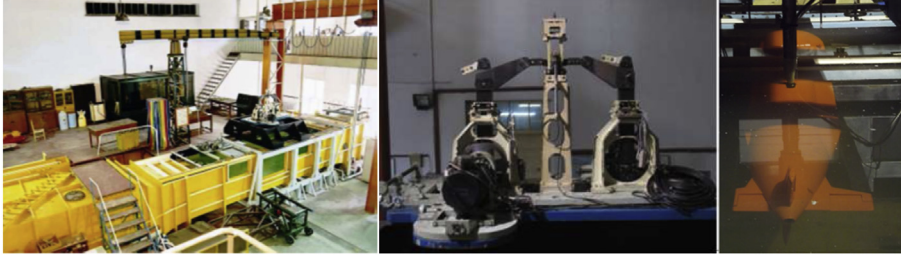


Fig. 11. Circulating water tunnel, PMM and submarine model under test conditions.

submarine model under test. Previous works by Yu and Su (2010) and Wang et al. (2015) have shown good experimental accuracy.

5.2. Model simplification

The HEU submarine is simulated in numerical environment using the spacial captive motion simulation method. Its hull form and CFD domain is as shown in Fig. 12. When employing the standard submarine equations, the hydrodynamic model also shows perfect fitting performance with values of *R-square* above 0.95 along all 6 degrees of freedom. Fig. 13 shows the *R-square* variations when simplifying hydrodynamic model of the HEU submarine model.

The variations of sway force and pitch moment are different from SUBOFF. Suddenly drop does not happen until *R-square* value below 0.90. This is mainly due to their geometry difference of sail and forward control surfaces. The deleted coefficients are Y'_p and Y'_r for sway force, and $M'_{w|w|}$, $M'_{w|w|}$, M'_{rp} and M'_w for pitch moment when *R-square* values drop between 0.95 and 0.90. Some of these coefficients cannot be determined using PMM test. The simplified hydrodynamic model with *R-square* value above 0.90 is shown as

$$X = \frac{1}{2}\rho l^3 (X'_{\dot{u}}\dot{u} + X'_{wq}wq) + \frac{1}{2}\rho l^2 X'_{uu}u^2 \quad (22)$$

$$Y = \frac{1}{2}\rho l^3 (Y'_{\dot{v}}\dot{v} + Y'_{p}up) + \frac{1}{2}\rho l^2 Y'_{uv}uv \quad (23)$$

$$Z = \frac{1}{2}\rho l^3 (Z'_{\dot{w}}\dot{w} + Z'_{q}uq) + \frac{1}{2}\rho l^2 Z'_{uw}uw \quad (24)$$

$$K = \frac{1}{2}\rho l^5 K'_{\dot{p}}\dot{p} + \frac{1}{2}\rho l^4 (K'_{p}up + K'_{\dot{v}}\dot{v}) + \frac{1}{2}\rho l^3 K'_{v}uv \quad (25)$$

$$M = \frac{1}{2}\rho l^5 M'_{\dot{q}}\dot{q} + \frac{1}{2}\rho l^4 (M'_{w\dot{w}}\dot{w} + M'_{q}uq) \quad (26)$$

$$N = \frac{1}{2}\rho l^5 N'_{\dot{r}}\dot{r} + \frac{1}{2}\rho l^4 (N'_{r}ur + \frac{1}{2}\rho l^3 N'_{v}uv) \quad (27)$$

For hydrodynamic model with *R-square* value above 0.95, only equations for sway force and pitch moment will be different, they can be written as.

$$Y = \frac{1}{2}\rho l^4 Y'_{\dot{p}}\dot{p} + \frac{1}{2}\rho l^3 (Y'_{\dot{v}}\dot{v} + Y'_{p}up + Y'_{r}ur) + \frac{1}{2}\rho l^2 Y'_{uv}uv \quad (28)$$

$$M = \frac{1}{2}\rho l^5 (M'_{\dot{q}}\dot{q} + M'_{rp}rp) + \frac{1}{2}\rho l^4 (M'_{w\dot{w}}\dot{w} + M'_{q}uq) + \frac{1}{2}\rho l^3 \left(M'_{uw}uw + M'_{w|w|}w \left| (v^2 + w^2)^{\frac{1}{2}} \right| + M'_{|w|u}|w| \right) \quad (29)$$

Table 5 demonstrates comparison of hydrodynamic coefficients between CFD and EFD results. The hydrodynamic coefficients determined by CFD approach are in good agreement with EFD result. Fig. 14 shows

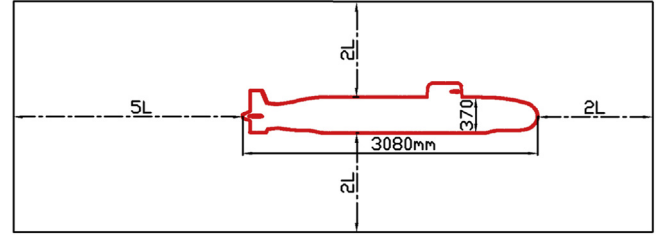


Fig. 12. Hull form and CFD domain.

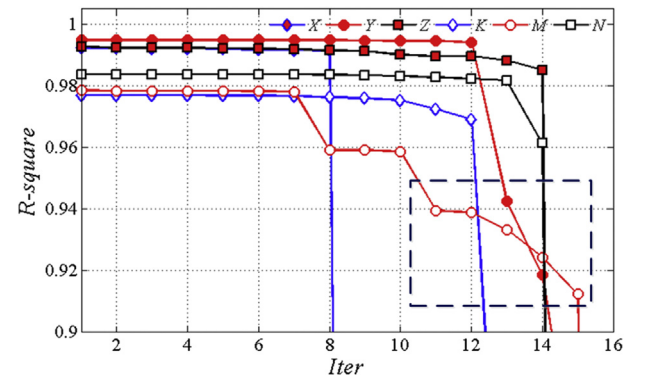


Fig. 13. *R-square* value variation with Equation simplification procedure for HEU submarine.

Table 5
Hydrodynamic coefficients of HEU submarine model.

Coefficients	CFD	EFD	Error
Y'_f	–	–0.000725	–
Y'_v	–0.021386	–0.021600	1.0%
Y'_r	–	0.007263	–
Y'_p	–0.001281	–	–
Y'_v	–0.045352	–0.053310	15.0%
Z'_q	–	–0.000251	–
Z'_w	–0.013140	–0.013200	0.5%
Z'_q	–0.009294	–0.010065	7.7%
Z'_w	–0.031663	–0.034633	8.6%
K'_p	–1.1E-5	–1.04E-5	–1.6%
K'_p	–0.000113	–0.000127	10.8%
K'_v	–0.000190	–	–
K'_v	–0.001181	–0.001050	–12.5%
M'_q	–0.000926	–0.000854	–8.4%
M'_w	–0.000390	–0.000502	22.4%
M'_q	–0.005359	–0.005307	–1.0%
M'_w	–	0.004780	–28.7%
N'_r	–0.001126	–0.001130	0.3%
N'_r	–0.005313	–0.005906	10.0%
N'_v	–	0.000664	–
N'_v	–0.011774	–0.016858	30.2%

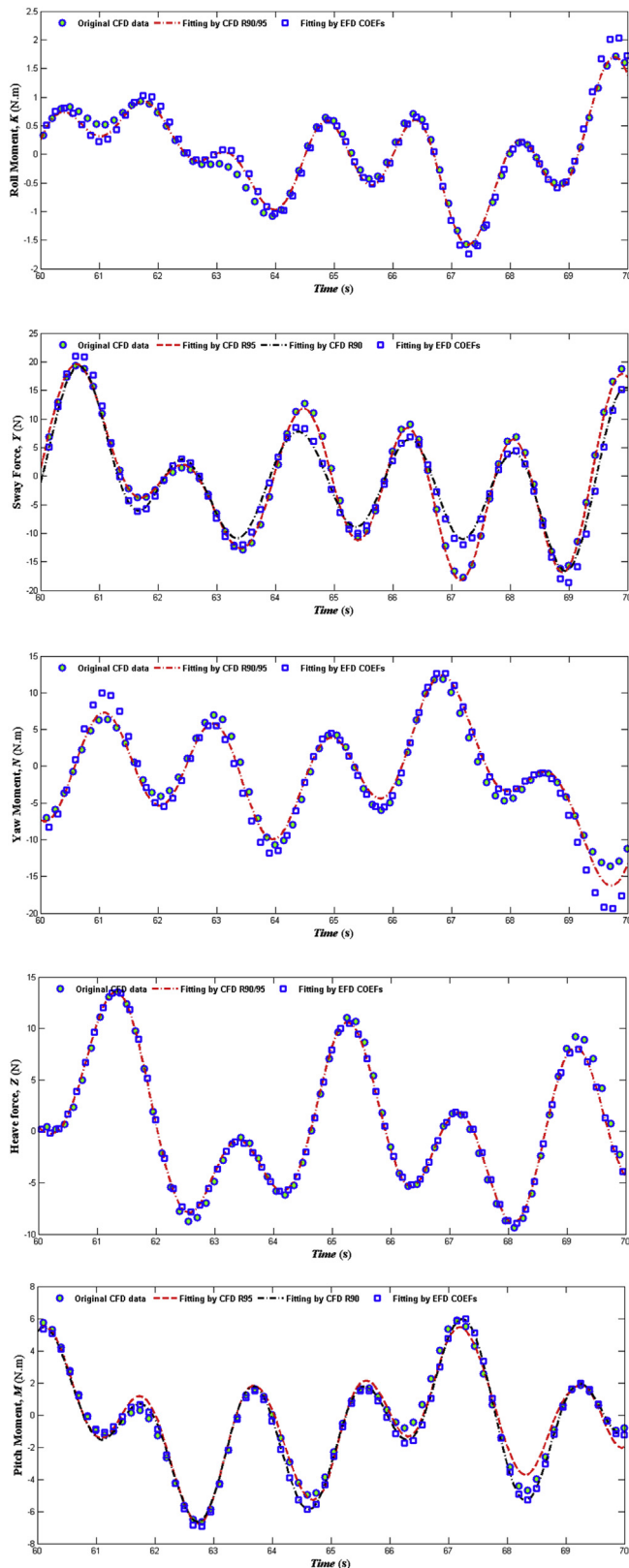


Fig. 14. Forces and Moments comparison between CFD result and fitting of CFD and EFD.

variation of forces and moments by CFD simulation, and their fitting curves according to hydrodynamic coefficients by CFD and EFD, respectively. Even though some coefficient errors in Table 5 are as large as 30%, their significance in hydrodynamic model is low, the simplified

hydrodynamic model with *R-square* value above 0.90 show good agreement with curves fitted by EFD coefficients. As to motion equations for sway force and pitch moment with *R-square* value above 0.95, they show better agreement with original CFD data. It indicates that those additional coefficients in equations (28) and (29) are necessary to better predict hydrodynamic performance of HEU submarine model, whereas they are not easy to be estimated by PMM or RA tests. The comparison results further prove that the spatial captive motion using CFD approach is efficient and reliable.

6. Conclusions

Model test, which expends a mass of labor cost, takes months to determine coefficients, even its numerical simulation takes weeks. They can only determine very limited coefficients, and repetitive tests or simulations have to be conducted. The developed spatial captive motion simulation method is very time-efficient, all hull-related coefficients can be determined by only one simulation, which can reduce time consumption to days, or even shorter when high accuracy is unnecessary for conceptual design.

Because the spatial captive motion simulation method can prove model validity easily, if fitting performance are verified, investigation on correlation and significance can help figure out which component shows so limited influence to the total forces that can be eliminated. The final simplified hydrodynamic model with much less coefficient while fitting accuracy remained. Targeted simplification of hydrodynamic model can be conducted for submarines with different geometric characteristics. Simplification results indicate that a majority of components in standard submarine motion equations are not quite necessary for geometry like SUBOFF and HEU submarine. Comparisons between curves fitted by CFD and EFD coefficients further prove the reliability of the proposed simulation and simplification method.

Spatial captive motion simulation not only can produce good predictions of hydrodynamic coefficients in motion equations, but also is appropriate for assessing reliability of a new hydrodynamic model. Besides investigating motions based on the steady state assumption, it is expected to have a broader scope in maneuverability research on strong maneuver or other nonlinear motions. Our next steps will include simulation accuracy improvement and investigation of body-rudder-propeller interaction. Accuracy could be improved by employing a better turbulence model and more computer resources. Overset mesh and actuator disk will be coupled into the simulation approach to determine rudder rotation and propeller related coefficients.

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